Lab Assignment 04

User Input and Advanced Loops

CSE110: Programming Language I

No of Tasks	Points to Score
15	**150**

Submit the coding tasks (Task 1 - 11) on buX and the handwritten tasks (Task 12 - 15) to your Lab Instructors in the beginning of the next lab class.

1. Write a Java program that asks the user how many inputs they want to provide and then takes that many inputs and prints the maximum, minimum, and average of all the **even positive numbers** given by the user. If no even positive number is given, the average should be zero.

Sample Input	**Sample Output**
5 12 -8 19 8 -1	Max: 12 Min: 8 Average: 10
Explanation: At first the user gave 5 as the input which indicates that the user will provide 5 numbers. Then 5 numbers were taken as inputs. Among these, only 12 and 8 are even positive numbers.	

2. Write a Java program that will keep taking integer numbers as inputs from the user and print the square of those numbers until it gets a negative number and then stop.
Sample Input/Output: (The purple numbers are input.)
Enter Number: 2
2 ^ 2 = 4
Enter Number: 6
6 ^ 2 = 36
Enter Number: 1
1 ^ 2 = 1
Enter Number: 4
4 ^ 2 = 16
Enter Number: -5

3. Write a Java code that asks an integer as input from the user and takes that many integer inputs. Your task is to count how many numbers are non-negative and negative.
Sample Input: (The purple numbers are input.)
Enter an integer: 9
Enter number 1: -8
Enter number 2: 33
Enter number 3: -100
Enter number 4: 10
Enter number 5: 0
Enter number 6: 5
Enter number 7: 10

Enter number 8: -4
Enter number 9: 4
Sample Output:
6 Non-negative Numbers
3 Negative Numbers

4. Write a Java program to take a positive integer ***N*** (where N > 0) as user input and print the **first *N* prime numbers starting from 2**. Your code should check all the positive integers starting from 2 and determine whether they are prime or not until ***N*** prime numbers are found.

Sample Input 1:
5
Sample Output 1:
2
3
5
7
11
Sample Input 2:
7
Sample Output 2:
2
3
5
7
11
13
17

5. Write a Java code of a program that reads the value of N (where N > 0) from the user and calculates the value of y if the expression of y is as follows:

$$y = -(1) - (1 + 2) - (1 + 2 + 3) - \ldots - (1 + 2 + 3 + \ldots + N)$$

Sample Input:
The value of N: 2
Sample Output:
The value of y: -4

Sample Input:

The value of N: 4
Sample Output:
The value of y: -20

6. Write a Java program that will keep taking even positive integer numbers as inputs from the user and print the number of divisors(**factors**) of those numbers until it gets an odd number and then stops.
Sample Input & Output: (The purple numbers are input)
Enter Number: 44
44 has 6 divisors
Enter Number: 30
30 has 8 divisors
Enter Number: 8
8 has 4 divisors
Enter Number: 4
4 has 3 divisors
Enter Number: 6
6 has 4 divisors
Enter Number: 20
20 has 6 divisors
Enter Number: 24
24 has 8 divisors
Enter Number: 5

7. Read an integer N that is the number of test cases that follow. Each test case contains two integers X and Y. Print one output line for each test case that the sum of Y odd numbers from X including it if is the case. For example:
For the input 4 5, the output must be 45, that is: 5 + 7 + 9 + 11 + 13
For the input 7 4, the output must be 40, that is: 7 + 9 + 11 + 13

Sample Input	**Sample Output**
2 4 3 11 2	21 24
Explanation: Here, the 2 means there are two test cases. For each test case you have to take two inputs (X, Y) and print the sum of Y odd numbers starting from X.	

8. Take the length and width of a **rectangle** from the user and create the rectangle according to the output below. Your output should match the specified output.

Sample Input #1 4 6	**Sample Input #2** 3 5
Output 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4 1 2 3 4	**Output** 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3

9. Take the height of a **right-justified right triangle** from the user and create the triangle according to the output below. Your output should match the specified output.

Sample Input #1 4	**Sample Input #2** 3

Output
  1
  1 2
  1 2 3
1 2 3 4
	Output
	  1
	  1 2
	1 2 3

10. Take the height of an **isosceles triangle** from the user and create the triangle according to the output below. Your output should match the specified output.

Sample Input #1 4	**Sample Input #2** 3

Output
  1
  1 2 3
  1 2 3 4 5
1 2 3 4 5 6 7
	Output
	  1
	  1 2 3
	1 2 3 4 5

11. Write a Java program that will ask for a range (a starting number and an ending number) from the user and print all the Armstrong numbers between that range.

*[**Armstrong Number:** An Armstrong number is a number whose sum of digits raised to the power the number of digits equals to that number.*

For example, 371 is an Armstrong number because $3^3 + 7^3 + 1^3 = 371$, here the total number of digits in 371 is 3 ***]***

<u>Sample Input 1:</u>
Start: 300
End: 500

<u>Sample Output 1:</u>
Armstrong numbers:
370
371
407
<u>Sample Input 2:</u>
Start: 100
End: 200
<u>Sample Output 2:</u>
Armstrong numbers:
153

12. Trace the following code, create a tracing table and write the outputs.

```
public class T1{
   public static void main(String args[]){
     int x = 0, y = 0;
     int sum = 0;
     while (x < 4){
       y = x - 3;
       while (y < 3){
           sum = (sum % 3) + x - y * 3 ;
           System.out.println(sum);
           y = y + 1;
       }
       if (x > 5){
           x++;
       }
       else{
           x += 2;
       }
     }
   }
}
```

13. Trace the following code, create a tracing table and write the outputs.

```
public class T2 {
  public static void main(String args[]) {
      int x = 0, i = 0, sum = 0;
      i = 1;
      x = 2;
      sum = 0;
      while (i < 20){
         x = x + i;
         sum = sum + x + 1;
         System.out.println(sum);
         if (x > 5){
           i += 2;
         }
         else {
           i += 3;
         }
      }
      sum = sum + i;
      System.out.println(sum);
   }
}
```

14. Trace the following code, create a tracing table and write the outputs.

```
public class T3
{
  public static void main(String args[])
  {
     int x = 0, y = 0;
     int sum = 0;
     while (x < 10){
       y = x - 3;
       y = 40;
       while (y > 22){
         if ((sum > 30) && (sum < 40)){
            sum = sum + x * 2 ;
         }
         else if ((sum > 40) && (sum < 50)){
            sum = sum + x * 3;
         }
         else {
            sum = sum + 23;
         }
         System.out.println(sum);
         y = y - 10;
       }
       x += 2;
     }
  }
}
```

15. Trace the following code, create a tracing table and write the outputs.

```
public class T4{
  public static void main(String args[]){
    boolean check = true;
    int x = 2, y = 2, z = 3;
    while(check){
          y = 4 / x % 3 + z * y - 5;
          if(y > 10 || x==7){
               z += 3;
               break;
          }
          if(4+x%3 > 5){
            x %= y + (z--) + z;
            System.out.println(x);
          }
          else{
            y += x + (--z) + y;
              System.out.println(y);
          }
          x++;
          System.out.println(x + y);
    }
  }
}
```